Quizrr Chapter-Wise Test for JEE Advanced - 2025

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Q.1

Let $f(x)$ be a differentiable function on $x \in R$ such that $f(x+y) = f(x) \cdot f(y)$ for all x, y. If $f(0) \neq 0,\ f(5) = 12$ and $f'(0) = 16$, then $f'(5)$ is equal to

(A) 190

(B) 186

(C) 196

(D) 192

Q.2

If $y = \frac{\sin^{-1} x}{\sqrt{1-x^2}}$, then $\left(1-x^2\right)\frac{dy}{dx}$ equals-

(A) $x + y$

(B) $1 + xy$

(C) $1 - xy$

(D) $xy - 2$

Q.3

If $y = \frac{1}{x^2-a^2}$, then $\frac{d^2y}{dx^2}$ equals-

(A) $\frac{3x^2+a^2}{(x^2-a^2)^3}$

(B) $\frac{3x^2+a^2}{(x^2-a^2)^4}$

(C) $\frac{2\left(3x^2+a^2\right)}{(x^2-a^2)^3}$

(D) $\frac{2\left(3x^2+a^2\right)}{(x^2-a^2)^4}$

Q.4

If the function $f(x) = -4e^{\frac{1-x}{2}} + 1 + x + \frac{x^2}{2} + \frac{x^3}{3}$ be an invertible function and $g(x) = f^{-1}(x)$, then the value of $g'\left(-\frac{7}{6}\right)$ equals:

(A) $\frac{1}{5}$

(B) $-\frac{1}{5}$

(C) $\frac{6}{7}$

(D) $-\frac{6}{7}$

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Q.5

If $x\sqrt{1+y}+y\sqrt{1+x}=0$, then $\frac{dy}{dx}$ equals-

(A) $\frac{1}{(1+x)^2}$

(B) $-\frac{1}{(1+x)^2}$

(C) $\frac{1}{1+x^2}$

(D) None of these

Q.6

If $y=\sin^{-1}\sqrt{\sin x}$, then $\frac{dy}{dx}$ equals-

(A) $\frac{2\sqrt{\sin x}}{\sqrt{1+\sin x}}$

(B) $\frac{\sqrt{\sin x}}{\sqrt{1-\sin x}}$

(C) $\frac{1}{2}\sqrt{1+\operatorname{cosec} x}$

(D) $\frac{1}{2}\sqrt{1-\operatorname{cosec} x}$

Q.7

If $y=\log_x 10$, then the value of dy/dx equals-

(A) $1/x$

(B) $10/x$

(C) $-\frac{(\log_x 10)^2}{x\log_e 10}$

(D) $\frac{1}{(x\log_e 10)}$

Q.8

If $\cos(xy)=x$, then $\frac{dy}{dx}$ is equal to -

(A) $\frac{y+\operatorname{cosec}(xy)}{x}$

(B) $\frac{y+\sin(xy)}{x}$

(C) $\frac{y+\cos(xy)}{x}$

(D) $-\frac{y+\operatorname{cosec}(xy)}{x}$

Q.9

Let $y=\sqrt{x+\sqrt{x+\sqrt{x+\ldots..\infty}}}$, then $\frac{dy}{dx}$ is equal to

(A) $\frac{1}{2y-1}$

(B) $\frac{x}{x+2y}$

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(C) $\frac{1}{\sqrt{1+4x}}$

(D) $\frac{y}{2x+y}$

Q.10

If $y=\frac{x^4-x^2+1}{x^2+\sqrt{3}x+1}$ and $\frac{dy}{dx}=ax+b$, then the value of $a-b$ is $-$

(A) $\cot\frac{\pi}{8}$

(B) $\cot\frac{5\pi}{12}$

(C) $\tan\frac{5\pi}{12}$

(D) $\tan\frac{5\pi}{8}$

Q.11

Choose the correct options -

(A) If $x^2e^y+2xye^x+13=0$, then $dy/dx=-\frac{2xe^{y-x}+2y(x+1)}{x(xe^{y-x}+2)}$

(B) If $x=a(\theta+\sin\theta), y=a(1-\cos\theta)$, then $dy/dx=\tan\frac{1}{2}\theta$

(C) If $y=\log\left(\frac{e^x}{e^x+1}\right)$, then $dy/dx=\frac{1}{e^x+1}$

(D) If $x^yy^x=1$, then $\frac{dy}{dx}=-\frac{y(y+x\log y)}{x(x+y\log x)}$

Q.12

Passage
$f(x)$ is a polynomial function $f:R\to R$ such that $f(2x)=f'(x)f''(x)$.
Question:
The value of f (3) is -

(A) 4

(B) 12

(C) 15

(D) None of these

Q.13

Passage
$f(x)$ is a polynomial function $f:R\to R$ such that $f(2x)=f'(x)f''(x)$.
Question:
f (x) is –

(A) one-one and onto

(B) one-one and into

(C) many-one and onto

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(D) many-one and into

Q.14

Passage

$f(x)$ is a polynomial function $f : R \to R$ such that $f(2x) = f'(x)f''(x)$.

Question:

Equation $f(x) = x$ has -

(A) three real and positive roots

(B) three real and negative roots

(C) one real roots

(D) three real roots such that sum of roots is zero

Q.15

Passage

$f(x) = x^2 + xg'(1) + g''(2)$ and $g(x) = f(1)x^2 + xf'(x) + f''(x)$.

Question:

The value of $f(3)$ is -

(A) 1

(B) 0

(C) -1

(D) -2

Q.16

Passage

$f(x) = x^2 + xg'(1) + g''(2)$ and $g(x) = f(1)x^2 + xf'(x) + f''(x)$.

Question:

The value of $g(0)$ is -

(A) 0

(B) -3

(C) 2

(D) None of these

Q.17

Passage

$f(x) = x^2 + xg'(1) + g''(2)$ and $g(x) = f(1)x^2 + xf'(x) + f''(x)$.

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Question:

The domain of the function $\sqrt{\frac{f(x)}{g(x)}}$ us -

(A) $(-\infty, 1] \cup (2, 3]$

(B) $(-2, 0] \cup (1, \infty)$

(C) $(-\infty, 0] \cup (2/3, 3]$

(D) None of these

Q.18

Match the column -

Column I	Column II
(A) $y = \sin^{-1}\left(\frac{2x}{1+x^2}\right)$, then $\frac{dy}{dx} = -\frac{2}{1+x^2}$	(p) for $x < 0$
(B) $y = \cos^{-1}\left(\frac{1}{\sqrt{1+x^2}}\right)$, then $\frac{dy}{dx} = \frac{-1}{1+x^2}$	(q) for $x > 1$
(C) $y = \left\|e^{\|x\|} - e\right\|$, then $\frac{dy}{dx} > 0$	(r) for $x < -1$
(D) $\mathrm{u} = \log\|2\mathrm{x}\|, \mathrm{v} = \left\|\tan^{-1}\mathrm{x}\right\|, then \mathrm{du/dv} > 2$	(s) for $-1 < x < 0$

(A) A-q,r ; B-p,r ; C-s,p ; D-p,s

(B) A-r,q ; B-p,q ; C-q,s ; D-q,r

(C) A-q,r ; B-p,r ; C-q,s ; D-q,r

(D) A-q,p ; B-p,q ; C-q,p ; D-r,s

Q.19

If $y = \left(1 + x^{1/4}\right)\left(1 + x^{1/2}\right)\left(1 - x^{1/4}\right)$, then $|dy/dx| =$

Q.20

If $y = \tan^{-1}\frac{3x - x^3}{1 - 3x^2}$, then $dy/dx = \frac{A}{1+x^2}$. Find the value of A.

Answers & Solutions

Q.1 Answer:

192

Solution:

$f'(5) = \lim_{h \to 0} \frac{f(5+h) - f(5)}{h}$

$= \lim_{h \to 0} \frac{f(5+h) - f(5+0)}{h}$

$= \lim_{h \to 0} \frac{f(5) \cdot f(h) - f(5) \cdot f(0)}{h}$

$[\because f(x+y) = f(x).\, f(y) \text{for all } x, y]$

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$= \left(\lim_{h\to 0} \frac{f(h)-f(0)}{h}\right) \cdot f(5)$

$f'(0) \times f(5) = 16 \times 12 = 192$

Q.2 Answer:

$1 + xy$

Solution:

From the given equation, we have $y^2\left(1-x^2\right) = \left(\sin^{-1} x\right)^2$

$\Rightarrow \quad \left(1-x^2\right) 2y\frac{dy}{dx} - 2xy^2 = 2\frac{\sin^{-1} x}{\sqrt{1-x^2}}$

$\Rightarrow \quad 2\left(1-x^2\right) y\frac{dy}{dx} - 2xy^2 = 2y \Rightarrow \left(1-x^2\right)\frac{dy}{dx} = 1 + xy$

Q.3 Answer:

$\frac{2\left(3x^2+a^2\right)}{\left(x^2-a^2\right)^3}$

Solution:

$\frac{dy}{dx} = \frac{-2x}{\left(x^2-a^2\right)^2} \Rightarrow \frac{d^2y}{dx^2} = -\frac{\left(x^2-a^2\right)^2 \cdot 2 - 2x \cdot 2\left(x^2-a^2\right)\cdot 2x}{\left(x^2-a^2\right)^4} = \frac{2\left(3x^2+a^2\right)}{\left(x^2-a^2\right)^3}$

Q.4 Answer:

$\frac{1}{5}$

Solution:

$\frac{dy}{dx} = \frac{4}{2}e^{\frac{1-x}{2}} + 1 + x + x^2 = 2e^{\left(\frac{1-x}{2}\right)} + x^2 + x + 1$

$g'(y) = \frac{dx}{dy} = \frac{1}{2e^{-\left(\frac{x-1}{2}\right)} + x^2 + x + 1}$

When $y = -\frac{7}{6}$ **then** $x = 1$

$\left.\frac{dy}{dx}\right]_{x=1, y=-7/6} = \frac{1}{2+3} = \frac{1}{5}$

Q.5 Answer:

$-\frac{1}{(1+x)^2}$

Solution:

Let us first express y in terms of x **because all alternatives are in terms of** x**.**

So $\mathrm{x}\sqrt{1+\mathrm{y}} = -\mathrm{y}\sqrt{1+\mathrm{x}}$

$\Rightarrow \quad x^2(1+y) = y^2(1+x)$

$\Rightarrow x^2 - y^2 + x^2y - y^2x = 0$

$\Rightarrow \quad (x-y)(x+y+xy) = 0$

$\Rightarrow \quad x + y + xy = 0$

$(x \neq y)$

$\Rightarrow \quad y = -\frac{x}{1+x}$

$\therefore \quad \frac{dy}{dx} = -\frac{(1+x)1 - x.1}{(1+x)^2} = -\frac{1}{(1+x)^2}$

Q.6 Answer:

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$\frac{1}{2}\sqrt{1+\operatorname{cosec} x}$

Solution:

$$\frac{dy}{dx}=\frac{1}{\sqrt{1-\sin x}}\cdot\frac{1}{2\sqrt{\sin x}}\cos x=\frac{\sqrt{1+\sin x}}{2\sqrt{\sin x}}=\frac{1}{2}\sqrt{1+\operatorname{cosec} x}$$

Q.7 Answer:

$-\frac{(\log_x 10)^2}{x\log_e 10}$

Solution:

$$y=\log_x 10=\frac{\log_e 10}{\log_e x}$$
$$\therefore\quad \frac{dy}{dx}=\log_e 10\left\{-\frac{1}{(\log_e x)^2}\cdot\frac{1}{x}\right\}=-\frac{1}{x\log_e 10}\frac{(\log_e 10)^2}{(\log_e x)^2}=-\frac{(\log_x 10)^2}{x\log_e 10}$$

Q.8 Answer:

$-\frac{y+\operatorname{cosec}(xy)}{x}$

Solution:

$$\cos(xy)-x=0$$
$$\therefore\quad \frac{dy}{dx}=-\frac{-y\sin(xy)-1}{-x\sin(xy)}=-\frac{y+\operatorname{cosec}(xy)}{x}$$

Q.9 Answer:

$\frac{1}{2y-1}$

Solution:

$$y^2=x+y\Rightarrow\frac{dy}{dx}=\frac{1}{2y-1}$$
$$\text{Also, } y=\frac{x}{y}+1\Rightarrow\frac{dy}{dx}=\frac{y}{2x+y}$$
$$\text{Also, } y^2-y-x=0\Rightarrow y=\frac{1\pm\sqrt{1+4x}}{2}$$
$$\Rightarrow\quad y=\frac{1+\sqrt{1+4x}}{2}.(\text{ as } y>0)$$

Q.10 Answer:

$\tan\frac{5\pi}{12}$

Solution:

$$y=\frac{\left(x^2+1\right)^2-3x^2}{x^2+\sqrt{3}x+1}=\frac{\left(x^2+1+\sqrt{3}x\right)\left(x^2+1-\sqrt{3}x\right)}{x^2+1+\sqrt{3}x}$$
$$\frac{dy}{dx}=2x-\sqrt{3}\Rightarrow a=2 \text{ and } b=-\sqrt{3}$$
$$a-b=2+\sqrt{3}=\tan\frac{5\pi}{12}=\cot\frac{\pi}{12}$$

Q.11 Answer:

If $x^2e^y+2xye^x+13=0$, then $dy/dx=-\frac{2xe^{y-x}+2y(x+1)}{x(xe^{y-x}+2)}$

Solution:

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(A) Let $\mathrm{f(x,y)=x^2e^y+2xye^x+13}$

$\therefore \quad \frac{dy}{dx}=-\frac{\partial f}{\partial x}/\frac{\partial f}{\partial y}=-\frac{2xe^y+2ye^x+2xye^x}{x^2e^y+2xe^x}$

Dividing Num ^{r} **and** Den^r **by** e^x

$\frac{dy}{dx}=-\frac{2xe^{y-x}+2y(x+1)}{x(xe^{y-x}+2)}$

(B) $\frac{\mathrm{dx}}{\mathrm{d}\theta}=\mathrm{a}(1+\cos\theta),\frac{\mathrm{dy}}{\mathrm{d}\theta}=\mathrm{a}\sin\theta$

$\therefore \quad \frac{dy}{d\theta}=\frac{dy/d\theta}{dx/d\theta}=\frac{a\sin\theta}{a(1+\cos\theta)}=\tan\frac{1}{2}\theta$

(C) $\mathrm{y=\log e^x-\log\left(e^x+1\right)=x-\log\left(e^x+1\right)}$

$\therefore \quad \frac{dy}{dx}=1-\frac{e^x}{e^x+1}=\frac{1}{e^x+1}$

(D) Taking log on both sides, we have $y\log x+x\log y=0$ **Now using partial derivatives, we have**

$\frac{dy}{dx}=-\frac{y/x+\log y}{\log x+x/y}=-\frac{y(y+x\log y)}{x(x+y\log x)}$

Q.12 Answer:

12

Solution:

Suppose degree of $\mathrm{f(x)=n}$**, then degree of** $\mathrm{f'=n-1}$ **and** $\deg \mathrm{f''=n-2}$**.**

So, $n=n-1+n-2$

Hence, $\mathrm{n=3}$

So put $f(x)=ax^3+bx^2+cx+d$ **(where** $a\neq 0$ **)**

From $\mathrm{f(2x)=f'(x)\cdot f''(x)}$

We have, $8ax^3+4bx^2+2cx+d=\left(3ax^2+2bx+c\right)(6ax+2b)$

$=18a^2x^3+18abx^2+\left(6ac+2b^2\right)x+2bc$

Comparing coefficients of terms, we have

$18a^2=8a \quad \Rightarrow a=4/9$

$18ab=4b\Rightarrow b=0$

$2c=6ac+4b^2\Rightarrow c=0$

$\mathrm{d=2bc \quad \Rightarrow d=0}$

$\Rightarrow \mathrm{f(x)=4x^3/9}$**, which is clearly one-one and onto**

$\Rightarrow \mathrm{f(3)=12}$

Also, $\frac{4\mathrm{x}^3}{9}=\mathrm{x}\Rightarrow \mathrm{x=0,x=\pm 3/2}$

Hence, sum of roots of equation is zero.

Q.13 Answer:

one-one and onto

Solution:

Suppose degree of $\mathrm{f(x)=n}$**, then degree of** $\mathrm{f'=n-1}$ **and** $\deg \mathrm{f''=n-2}$**.**

So, $\mathrm{n=n-1+n-2}$

Hence, $\mathrm{n=3}$

So put $f(x)=ax^3+bx^2+cx+d$**(where** $a\neq 0$**)**

From $f(2x)=f'(x)\cdot f''(x)$

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$\Rightarrow f(x) = 4x^3/9$, **which is clearly one-one and onto**

$\Rightarrow f(3) = 12$

Also, $\frac{4x^3}{9} = x \Rightarrow x = 0, x = \pm 3/2.$

Hence, sum of roots of equation is zero.

Q.14 Answer:

three real roots such that sum of roots is

zero

Solution:

Suppose degree of $f(x) = n$**, then degree of** $f' = n - 1$ **and** $\deg f'' = n - 2.$

So, $n = n - 1 + n - 2$

Hence, $n = 3$

So put $f(x) = ax^3 + bx^2 + cx + d$**(** **where** $a \neq 0$**)**

From $f(2x) = f'(x) \cdot f''(x)$

We have, $8ax^3 + 4bx^2 + 2cx + d = \left(3ax^2 + 2bx + c\right)(6ax + 2b)$

$= 18a^2x^3 + 18abx^2 + \left(6ac + 2b^2\right)x + 2bc$

Comparing coefficients of terms, we have

$18a^2 = 8a \Rightarrow a = 4/9$

$18ab = 4b \Rightarrow b = 0$

$2c = 6ac + 4b^2 \Rightarrow c = 0$

$d = 2bc \quad \Rightarrow d = 0$

$\Rightarrow f(x) = 4x^3/9$, **which is clearly one-one and onto**

$\Rightarrow f(3) = 12$

Also, $\frac{4x^3}{9} = x \Rightarrow x = 0, x = \pm 3/2.$

Hence, sum of roots of equation is zero.

Q.15 Answer:

0

Solution:

Here put g ' $(1) = a, g''(2) = b$

Then $f(x) = ax^2 + ax + b, f(1) = 1 + a + b \Rightarrow f'(x) = 2x + a, f''(x) = 2$

$\therefore \quad g(x) = (1 + a + b)x^2 + (2x + a)x + 2 = x^2(3 + a + b) + ax + 2$

$\Rightarrow g'(x) = 2x(3 + a + b) + a$ **and** $g''(x) = 2(3 + a + b)$

Hence, $g'(1) = 2(3 + a + b) + a$

$g''(2) = 2(3 + a + b)$

From (1), (2) and (3), we have $a = 2(3 + a + b) + a$ **and** $b = 2(3 + a + b)$

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$\Rightarrow 3+a+b=0$ and $b+2a+6=0$

Hence, $b=0$ and $a=-3$.

So, $f(x)=x^2-3x$ and $g(x)=-3x+2$

$\sqrt{\frac{f(x)}{g(x)}}=\sqrt{\frac{x^2-3x}{-3x+2}}$ is defined if $\frac{x^2-3x}{-3x+2}\geq 0$

$\Rightarrow \quad \frac{x(x-3)}{(x-2/3)}\leq 0 \Rightarrow x\in(-\infty,0]\cup(2/3,3]$

Q.16 Answer:

2

Solution:

Here put $g'(1)=a, g''(2)=b$

Then $f(x)=x^2+ax+b, f(1)=1+a+b \Rightarrow f'(x)=2x+a, f''(x)=2$

$g(x)=(1+a+b)x^2+(2x+a)x+2=x^2(3+a+b)+ax+2$

$g'(x)=2x(3+a+b)+a$ and $g''(x)=2(3+a+b)$

Hence, $g'(1)=2(3+a+b)+a$

$g''(2)=2(3+a+b)$

From $(1),(2)$ and (3), we have $a=2(3+a+b)+a$ and $b=2(3+a+b)$

$3+a+b=0$ and $b+2a+6=0$

Hence, $b=0$ and $a=-3$.

So, $f(x)=x^2-3x$ and $g(x)=-3x+2$

$f(0)=0$

$g(0)=2$

$\sqrt{\frac{f(x)}{g(x)}}=\sqrt{\frac{x^2-3x}{-3x+2}}$ is defined if $\frac{x^2-3x}{-3x+2}\geq 0$

$\frac{x(x-3)}{(x-2/3)}\leq 0 \Rightarrow x\in(-\infty,0]\cup(2/3,3]$

Q.17 Answer:

$(-\infty,0]\cup(2/3,3]$

Solution:

Here put $g'(1)=a, g''(2)=b$

Then $f(x)=ax^2+ax+b, f(1)=1+a+b \Rightarrow f'(x)=2x+a, f''(x)=2$

$g(x)=(1+a+b)x^2+(2x+a)x+2=x^2(3+a+b)+ax+2$

$g'(x)=2x(3+a+b)+a$ and $g''(x)=2(3+a+b)$

Hence, $g'(1)=2(3+a+b)+a$

$g''(2)=2(3+a+b)$

From $(1),(2)$ and (3), we have $a=2(3+a+b)+a$ and $b=2(3+a+b)$

$3+a+b=0$ and $b+2a+6=0$

Hence, $b=0$ and $a=-3$.

So, $f(x)=x^2-3x$ and $g(x)=-3x+2$

$\sqrt{\frac{f(x)}{g(x)}}=\sqrt{\frac{x^2-3x}{-3x+2}}$ is defined if $\frac{x^2-3x}{-3x+2}\geq 0$

$\frac{x(x-3)}{(x-2/3)}\leq 0 \Rightarrow x\in(-\infty,0]\cup(2/3,3]$

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Q.18 Answer:

A-q,r ; B-p,r ; C-q,s ; D-q,r

Solution:

(A) - qr, (B) - pr, (C) qs, (D) - qr. (A) We know that $2\tan^{-1}x =$

$$\begin{cases} \sin^{-1}\left(\frac{2x}{1+x^2}\right), & \text{if } -1 \le x \le 1 \\ \pi - \sin^{-1}\left(\frac{2x}{1+x^2}\right), & \text{if } x > 1 \\ -\pi - \sin^{-1}\left(\frac{2x}{1+x^2}\right), & \text{if } x < -1 \end{cases}$$

$\Rightarrow \quad \frac{dy}{dx} = -\frac{2}{1+x^2}$ **if** $x < -1$ **or** $x > 1$

(B) $\cos^{-1}\left(\frac{1}{\sqrt{1+x^2}}\right) = \begin{cases} \tan^{-1}x, x \ge 0 \\ -\tan^{-1}x, x < 0 \end{cases}$

$\Rightarrow \quad \frac{dy}{dx} = -\frac{1}{1+x^2}$ **if** $x < 0$

(C) $\Rightarrow \frac{dy}{dx} > 0$ **if** $x > 1$ **or** $-1 < x < 0$

(D) $u = \log|2x|, v = \left|\tan^{-1}x\right|$

$$\Rightarrow \frac{du}{dx} = \frac{1}{x} \text{ and } \frac{dv}{dx} = \begin{cases} \frac{1}{1+x_1^2}, & x > 0 \\ -\frac{1}{1+x^2}, & x < 0 \end{cases}$$

$$\Rightarrow \frac{du}{dv} = \begin{cases} \frac{1+x^2}{x}, x > 0 \\ -\frac{1+x^2}{x}, x < 0 \end{cases}$$

$\Rightarrow$ **if** $x < -1$ **or** $x > 1$

Q.19 Answer:

—

Solution:

$$y = \left(1 + x^{1/2}\right)\left(1 - x^{1/2}\right) = 1 - x$$

$$\therefore \quad \frac{dy}{dx} = -1$$

Q.20 Answer:

—

Solution:

$$y = \tan^{-1}\frac{3x - x^3}{1 - 3x^2} = 3\tan^{-1}x$$

$$\therefore \quad \frac{dy}{dx} = \frac{3}{1+x^2}$$

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